

Investigating a function with the derivative

Lessons 19–22 · Algebra 11, §1.7 · P1 · 7.7–7.8 · 28 problems

Solve $y' = 0$ for the stationary points, then classify each one — by the sign of y' on either side, or by y'' . Always put the x -value back into y , not into y' .

NAME _____

CLASS _____

DATE _____

Easy 7

- | | |
|---|---|
| 1 Stationary point of $y = x^2 - 8x$ | 2 Classify it |
| 3 Stationary point of $y = 6x - x^2$, and its nature | 4 Is $y = 5 - 2x$ increasing or decreasing? |
| 5 Find y'' for $y = x^4 - 2x^2$ | 6 Where is $y = x^2 - 2x$ increasing? |
| 7 Stationary point of $y = x^3 + 1$, and its nature | |

Medium 7

- | | |
|---|--|
| 8 Investigate $y = x^3 - 6x$ | 9 Investigate $y = 2x^3 - 9x^2 + 12x$ |
| 10 Where is $y = x^3 - 3x^2$ decreasing? | 11 Stationary points of $y = x^4 - 4x^3$ |
| 12 Give the nature of each | 13 Point of inflection of $y = x^3 - 9x^2$ |
| 14 Greatest value of $y = x^3 - 3x$ on $[0, 2]$ | |

Hard 7

- 15 Investigate $y = \frac{x^2}{x^2 + 3}$
- 16 Find a so that $y = x^3 - ax$ has stationary points at $x = \pm 2$.
- 17 Find a and b so that $y = x^3 + ax^2 + bx$ has a stationary point at $(1, -2)$.
- 18 Show that $y = x^3 + 4x - 1$ is increasing for every x .
- 19 Find the intervals on which $y = 3x^4 - 4x^3 - 12x^2$ is increasing.
- 20 $y = x^4 + kx^2$ has exactly one stationary point. Find the values of k .
- 21 Find and classify the stationary points of $y = x^3 - 3x^2 - 9x + 5$.

Very hard 7

- 22 Show that $y = x^4$ has a minimum at $x = 0$ even though $y''(0) = 0$, and explain why the second-derivative test fails.
- 23 Investigate $y = (x - 1)^2(x + 2)$ fully: stationary points, their nature, and the point of inflection.
- 24 Find the values of k for which $y = x^3 + 3x^2 + kx$ has no stationary point.
- 25 $f(x) = ax^3 + bx^2 + cx$ has a maximum at $x = -1$, a minimum at $x = 3$ and $f(-1) = 10$. Find a , b and c .

26 The curve $y = x^3 + px + q$ has a stationary point at $(2, -6)$. Find p and q , and the other stationary point.

27 Find the greatest and least values of $y = x^3 - 6x^2 + 9x + 1$ on $[0, 4]$.

28 Show that if $f'(a) = 0$ and $f''(a) > 0$ then f has a local minimum at a .
